

# CdS photoconductive cell Resin coating type (5R type)

Standard type designed for general-purpose, wide application



CdS photoconductive cells utilize photoconductive effects in semiconductors that decrease their resistance when illuminated by light. These sensors are non-polar resistive elements with spectral response characteristics close to the human eye (luminous efficiency), thus making their operating circuits simple and small.

## Features

- Small size, thin package
- Low price
- Wide range of sensor lineup

## Applications

- Programed electronic shutter and stroboscope light control for compact camera
- Auto dimmer for digital display, CRT and room illumination
- Sensor for automatic light on/off
- Sensor for electronic toy and teaching aid material

### ■ Absolute maximum ratings / Characteristics (Typ. Ta=25 °C, unless otherwise noted)

| Type No. | Absolute maximum ratings |                     |                        | Characteristics *1             |                    |      |         |                                   |                        |              |
|----------|--------------------------|---------------------|------------------------|--------------------------------|--------------------|------|---------|-----------------------------------|------------------------|--------------|
|          | Supply voltage           | Power dissipation P | Ambient temperature Ta | Peak sensitivity wavelength λp | Cell resistance *2 |      |         | γ <sup>100</sup> <sub>10</sub> *4 | Response time 10 lx *5 |              |
|          |                          |                     |                        |                                | 10 lx, 2856 K      |      | 0 lx *3 |                                   | Rise time tr           | Fall time tf |
|          |                          |                     |                        |                                | Min.               | Max. | Min.    |                                   |                        |              |
|          |                          |                     |                        |                                | (Vdc)              | (mW) | (°C)    | (nm)                              | (kΩ)                   | (kΩ)         |
| P687-02  | 100                      | 30                  | -30 to +50             | 620                            | 5                  | 20   | 5.0     | 0.70                              | 60                     | 25           |
| P1201-04 |                          | 50                  | -30 to +60             | 540                            | 50                 | 200  | 20      | 0.90                              | 40                     | 30           |
| P1201-06 |                          |                     |                        |                                |                    | 100  |         |                                   |                        |              |
| P1241-04 | 50                       | 30                  | -30 to +60             | 560                            | 3                  | 9    | 0.2     | 0.70                              | 50                     | 40           |
| P1241-05 |                          |                     |                        |                                | 8                  | 24   | 0.5     | 0.70                              |                        |              |
| P1241-06 |                          |                     |                        |                                | 5                  | 20   | 0.5     | 0.75                              |                        |              |
| P1444    | 100                      |                     | -30 to +50             | 620                            | 10                 | 50   | 5.0     | 0.85                              | 40                     | 10           |
| P1445    |                          |                     |                        |                                | 48                 | 140  | 20      |                                   |                        |              |

\*1: All characteristics are measured after exposure to light (100 to 500 lx) for one to two hours.

\*2: The light source is a standard tungsten lamp operated at a color temperature of 2856 K.

\*3: Measured 10 seconds after shutting off the 10 lx light.

\*4: Typical gamma characteristics (within ±0.10 variations) between 100 lx to 10 lx

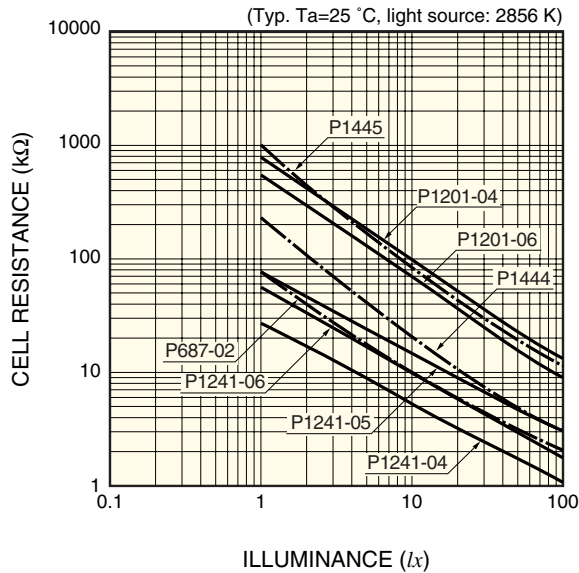
$$\gamma_{10}^{100} = \frac{|\log(R_{100}) - \log(R_{10})|}{|\log(E_{100}) - \log(E_{10})|}$$

E<sub>100</sub>, E<sub>10</sub>: illuminance 100 lx, 10 lx

R<sub>100</sub>, R<sub>10</sub>: resistance at 100 lx and 10 lx respectively

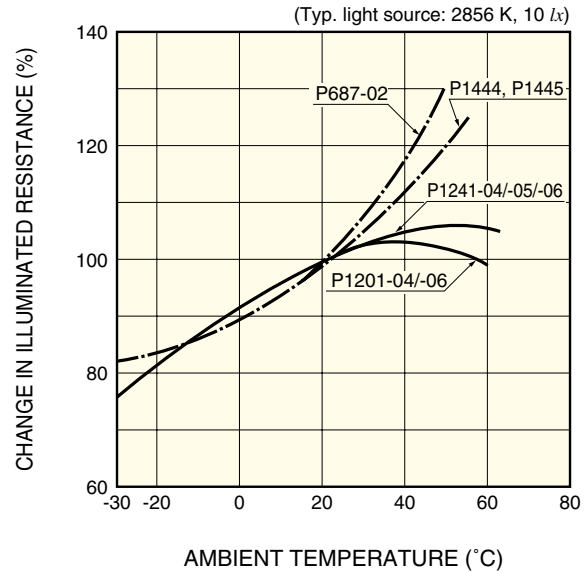
\*5: The rise time is the time required for the sensor resistance to reach 63 % of the saturated conductance level (when fully illuminated). The fall time is the time required for the sensor resistance to decay from the saturated conductance level to 37 %.

## Cell resistance vs. illuminance



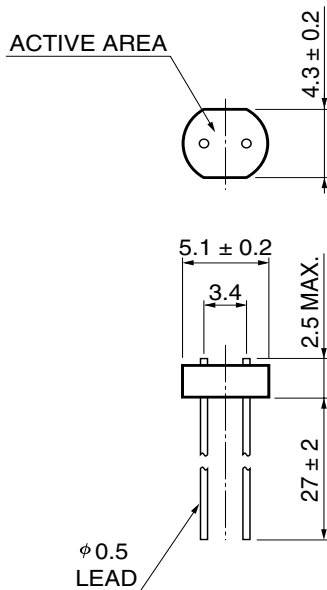
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## Resistance temperature characteristics



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## Dimensional outline (unit: mm)



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